

Part B:

Question No. 1:

Problem Analysis Chart (PAC):

Input	Processing	Output	Decision
• N (Total guest)	loops N line	• guest price (final price per guest)	• Season check (off season - on season)
• Season (peak/off peak)	nightly rate based on season and room type	• Total revenue (Hostel's total revenue)	• Room type check (standard, Deluxe, suite)
• Room type (standard / Deluxe / suite)	Computer base cost. Base cost = rate $\times$ night		• long-stay check nights $> 7 \Rightarrow 15\%$ th
• nights (no of nights)	• Apply 15% disc if night > 7		•
	• Calculate final guest price		
	• Accumulate into total revenue		

IPO Chart:

Input	Processing	Output
• N • season • room type • nights	1. Initialize total revenue = 0 2. Prompt user for number of guest N 3) For each guest from 1 to N. → a) Read season, room type, nights → b) Determine nightly rate using nested condition/switch statement → c) Calculate base price = rate $\times$ night → d) if night > 7 then discount = base price $\times 0.15$, ELSE discount = 0. → e) Calculate Total price = Base price - Discount → f) Add Total guest Price to Total Revenue → g) Output Total Price 4) Output: Total Revenue.	• guest Price • Total Revenue

Algorithm:

- 1) Start
- 2) Initialize Total Revenue = 0
- 3) Input total number of guest, N.
- 4) Set counter $i = 1$.
- 5) Repeat while $i \leq N$
 - Input season, room type, and nights.
 - if season is "Peak":
 - if room type is 'Standard', set rate = 5000.
 - Else if room type is 'Delux', set rate = 8000.
 - Else if room type is 'Suite', set rate = 12000.
 - Else (if season is 'Off-Peak')
 - if room type is "Standard", set rate = 3000.
 - Else if room type is 'Deluxe', set rate = 5000.
 - Else if room type is "Suite", set rate = 8000.
 - Compute base price = rate * nights.
 - if night > 7
 - Compute discount = base price * 0.15
 - Else
 - Compute discount = 0
 - Compute guest Price = base Price - discount
 - Add guest Price to Total Revenue
(Total Revenue = Total Revenue + guest Price)
 - Display guest Price
 - Increment $i = i + 1$
- 6) Display Total Revenue
- 7) End.

Pseudocode:

Begin

Declare N, nights, counter as INTEGER

Declare rate, basePrice, discount, guestPrice, totalRevenue REAL

Declare season, roomType as STRING

SET totalRevenue = 0

Print "Enter total number of guest"

Read N

FOR counter = 1 to N DO

Print Enter season (Peak/Off-Peak)

Read season

Print Enter room type (Standard/Deluxe/Suite);

Read roomType

Print "Enter number of nights stayed"

Read nights

If season == "Peak" Then

If roomType == "Standard" Then rate = 5000

Else if roomType == "Deluxe" Then

rate = 8000

Else if roomType == "Suite" Then

rate = 12000

END if

Else

if roomType == "Standard" Then

```
rate = 3000
Else if roomType == "Delux" Then
    rate = 5000
Else if roomType == "Suite" Then
    rate = 8000
ENDIF
ENDIF
```

base Price = rate * nights

```
If nights > 7 Then
    discount = base Price * 0.15
```

Else

discount = 0

```
ENDIF if
```

guest Price = base Price - discount

Total Revenue = Total Revenue + guest Price

```
Print "Final Price for Guest", counter, "is Rs,"
ENDIF FOR
```

Print "Hotel Total Revenue is," Total Revenue

```
ENDIF
```

flow chart:

